

Zhongyuan Hu

Research Intern | Generative AI · Video Generation · 3D Human Motion · Multimodal Learning

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PROFILE

Direct-entry Ph.D. student at Tsinghua University with three accepted first/co-first-author papers, including ECCV 2026 and ACM MM 2026, and research deployed at Tencent Games. Research focuses on controllable generative models for human motion, camera trajectories, video, and multimodal signals.

TECHNICAL EXPERTISE

Research: Generative AI, 3D Human Motion Generation, Video Generation, Multimodal Learning

Methods: Diffusion Models, LLMs, Retrieval-Augmented Generation, Mixture-of-Experts, Flow Matching, Camera Trajectory Generation, 3D Human Reconstruction

EDUCATION

Tsinghua University — Ph.D. in Control Science and Engineering

Sep 2025 - Jun 2030 (Expected)

Direct-entry Ph.D. student; advised by Prof. Xiu Li.

Chongqing University — B.Eng. in Computer Science and Technology

Sep 2021 - Jun 2025

GPA 3.91/4.00; ranked 1/323 (top 0.3%); National Scholarship (top 1%); First-Class Scholarship (top 3%).

INDUSTRY RESEARCH EXPERIENCE

Tencent Games — Research Intern, Shenzhen

Jun 2025 - Dec 2025

WATCH: World-aware Allied Trajectory and pose reconstruction for Camera and Human — Co-first Author, Core Contributor, Accepted

- Developed heading-angle decomposition and global-velocity modeling that explicitly leverage camera rotation and translation, reducing long-horizon drift in in-the-wild monocular human reconstruction.
- Outperformed GVHMR and WHAM; replacing GVHMR's corresponding module further improved reconstruction under dynamic cameras. The resulting algorithm was deployed in Tencent products.

arxiv.org/abs/2509.04600

SELECTED RESEARCH & PUBLICATIONS

Camera Operator: Object-Grounded Camera Trajectory Generation from Text and 3D Bounding Box Sequences

Jan 2026 - Apr 2026

First Author, Core Contributor — ACM MM 2026

- Built BlockCam, a 41K-sequence benchmark combining Unreal Engine and real-video pipelines with aligned text, 3D bounding-box state sequences, and camera trajectories.
- Proposed a geometry-conditioned flow-matching model with look-at, framing, visibility, and temporal supervision to preserve stable long-horizon camera-subject interaction.

doi.org/10.1145/3767308.3835459

InfiniteDance: Scalable 3D Dance Generation Towards in-the-wild Generalization

Mar 2025 - Aug 2025

Co-first Author, Core Contributor — ECCV 2026

- Developed an automated monocular motion-reconstruction pipeline and a foot-restoration diffusion model, producing over 100 hours of high-quality multimodal 3D dance data.
- Co-developed ChoreoLLaMA with RAG for unseen music and a slow-fast Mixture-of-Experts module for tempo-adaptive motion generation.

arxiv.org/abs/2603.13375 | infinitedance.github.io

ADDITIONAL PUBLICATIONS

Controllable Video Generation: A Survey — Co-first Author, Core Contributor; ACM Computing Surveys, under review.

arxiv.org/abs/2507.16869

AToM: Aligning Text-to-Motion Model at Event-Level with GPT-4Vision Reward — Co-author; CVPR 2025.

arxiv.org/abs/2411.18654